A064 v0.1.0: PKLSA Geometric Holonomy and Interference Test

SST-Workbench: independent numerical pilot

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1 Scientific scope

This work tests geometric Gauss linking and ideal filament circulation using four SHA-256-registered KnotPlot-native PKLSA carriers. It does not derive an SST gauge field, quantum representation, dynamical matter motion, or self-confinement. The admission gate remains closed because the available E010 v0.3.1 geometry layer is not publication-ready and all four samples originate from one provider.

2 Geometry and circulation

For oriented disjoint closed curves $C_i, C_j \subset \mathbb{R}^3$,

$$\mathrm{Lk}(C_i, C_j) = \frac{1}{4\pi} \oint_{C_i} \oint_{C_j} \frac{(d\mathbf{r}_i \times d\mathbf{r}_j) \cdot (\mathbf{r}_i - \mathbf{r}_j)}{\|\mathbf{r}_i - \mathbf{r}_j\|^3}. \tag{1}$$

The Biot–Savart velocity generated by a filament C_j at circulation Γ [$\mathrm{m^2\,s^{-1}}$] obeys

$$\mathbf{u}_j(\mathbf{x}) = \frac{\Gamma}{4\pi} \oint_{C_j} \frac{d\mathbf{r}_j \times (\mathbf{x} - \mathbf{r}_j)}{\|\mathbf{x} - \mathbf{r}_j\|^3}, \quad \oint_{C_i} \mathbf{u}_j \cdot d\boldsymbol{\ell} = \Gamma\, \mathrm{Lk}(C_i, C_j). \tag{2}$$

The normalized circulation is dimensionless. When KnotPlot coordinates are unitless, no measured SI conversion may be inferred. A finite regulator a replaces r^{-3} by $(r^2 + a^2)^{-3/2}$; this regulates the field but does not preserve exact topological quantization at finite a.

3 Model-only interference control

The *assigned*, not derived, $\mathbb{Z}_2$ representation $q = 1/2$ yields

$$\phi_{ij} = 2\pi q\, \mathrm{Lk}(C_i, C_j), \quad P_{\mathrm{toy}} = \frac{1 + \cos\phi_{ij}}{2}. \tag{3}$$

In particular, integer linking parity forces the toy-model probabilities to 0 and 1. This equation is an *analytic code benchmark*, not a prediction of SST-mediated propagation.

4 Gates

G0: exact geometry SHA-256, component counts, periodic closure, finite inputs. G1: linking integer residual < 0.035. G2: quadrature order-2 vs order-4 residual < 0.035. G3: orientation, basepoint and translation controls. G4: assigned toy-phase arithmetic and synthetic Hopf/unlink controls. G5–G10: independent providers, dynamical phase connection, physical propagation, higher-order three-component invariants, framework v1.0.7 integration and production geometry qualification remain unresolved until independently verified.

5 Falsifying experiment

Specify the finite-core incompressible Euler velocity, core law, Kelvin-wave spectrum, initial phase and a unique $\mathbf{A}_{\mathrm{eff}}\,[\mathrm{m}^{-1}]$ prior to selecting any particle label or observed flux. Compare predicted transfer of excitations over time against zero-connection, randomized phase, geometric reparameterizations and multiple independent providers. Report failure even if an unrelated gate passes.

References

[1] S. Saner et al., "Aharonov–Bohm interference in a $\mathbb{Z}_2$ lattice gauge theory on a hybrid qubit–oscillator quantum computer," *Nature Physics* (2026), https://doi.org/10.1038/s41567-026-03400-6.

[2] C. F. Gauss, "Zur mathematischen Theorie der electrodynamischen Wirkungen," unpublished manuscript (1833), in *Werke*, vol. 5 (1867), 605–629.

[3] H. K. Moffatt, "The degree of knottedness of tangled vortex lines," *Journal of Fluid Mechanics* **35**, 117–129 (1969), https://doi.org/10.1017/S0022112069000991.